

ing Gerber and Excellon. This software also allows users to directly import compressed zip files. Its salient features include high-resolution picture export, integrated screen-shot function, multiple draw modes, import and display of pictures behind the real CAD data, and multiple selection possibilities including transparent drawing.

Users can move, rotate and mirror layers/objects. The application framework with the software reads/writes Spectra file format and uses a plug-in style of interface for router engines.

AnadigmDesigner

AnadigmDesigner EDA software lets users develop circuits using a simple drag-and-drop solution. The software enables users to design and implement dynamically reconfigurable analogue circuits easily. It provides configurable analogue modules (CAMs), which users can drag and drop to develop their circuit. Each of these CAMs can be used to implement a range of analogue functions for which users set the parameters.

AnadigmAssistant tools included with AnadigmDesigner take the design process to a higher level of abstraction, fully automating the construction of common filter types and proportional integral, derivative (PID) controller circuits. AnadigmDesigner includes a time-domain functional simulator that provides a convenient way to assess the circuit's behaviour without the need for a lab set-up. The simulator's user interface is intuitive and easy to learn. Most of the steps are similar to bench testing. Using the software users can build a complete analogue system rapidly, simulate it immediately, then just point-and-click to download it to field-programmable analogue array (FPGA) chip for testing and validation.

AnadigmDesigner lets users develop designs using FPAs that can be reconfigured by the MCU in real time to change the function they perform within a system or to adapt on-the-fly to maintain precision. AnadigmDesigner also automatically translates a design into C code, allowing the design to be adjusted and controlled by a microprocessor within an embedded system.

Winlog Lite

Winlog Lite is a SCADA/HMI software package for supervision of industrial and civil plants. The integrated development environment provides different tools like gate builder, template builder and code builder for easy and intuitive creation of multi-language applications. The software comes with an extensive library of drivers and an OPC client interface for communication with electronic devices such as PLCs, controllers and motor drives.

Winlog Lite makes it possible to set up a distributed client/server architecture with TCP/IP protocol on intranet/Internet network or create Web applications accessible from standard browsers. Development tools include symbol factory and integrated editor. Symbol factory is a library of automation graphic objects with over 4000 manufacturing and industrial objects such as pumps, valves, motors, tanks, PLCs, piping and ISA symbols. The integrated editor allows resizing and change in colour, scheme and orientation of objects.

The software enables reports and historical data to be recorded in a standard format (DBF, CSV) that can be easily accessed by other Windows applications (Excel, Access, etc). Each menu item, template, tag, etc can be protected by the specification of the groups authorised to access it and those who can modify it. A name, password, and membership of one or more groups can then be defined for an unlimited number of operators. Every operator's action that has produced a data modification is recorded in a history file by date, time, event description and operator's name.

The software comes in two editions: full mode and demo mode. While it is necessary to purchase a licence for full mode, users can use the software without any registration in demo mode. In demo mode, communication with external peripherals stops after 15 minutes, and they have to be manually restarted.

Scilab

Scilab is a free and Open Source software for numerical computation providing a powerful computing environment for engineering and scientific applications. It

provides users a numerical environment similar to MATLAB. It has a high-level programming language allowing access to advanced data structures, 2D and 3D graphical functions, and lots of in-built mathematical functions.

Functionalities offered by Scilab include maths and simulation for usual engineering and science applications including mathematical operations and data analysis. It also offers 2D and 3D visualisation functions to visualise, annotate and export data, and many ways to create and customise various types of plots and charts.

It also offers algorithms to solve constrained and unconstrained continuous and discrete optimisation problems, tools to perform data analysis and modeling, control system design and analysis functions. Other important tools include signal processing, application development and Xcos simulator.

Due to its ability to interconnect with third-party technologies and applications, Scilab can also act as a unique platform to bring together codes written in different programming languages in a single, unified language, thus facilitating their distribution, back-up and use.

Scilab can interact directly with other languages such as Java, Python, Tcl Tk, Fortran, C and C++. These languages can be called directly from within Scilab. Scilab can also be used as a standalone calculation engine by Java, Python, C or C++ external applications.

GNU Octave

GNU Octave is a high-level language, primarily intended for numerical computations. It provides a command-line interface for solving linear and non-linear problems numerically. It may also be used as a batch-oriented language.

Octave has extensive tools for solving common numerical linear algebra problems, finding the roots of non-linear equations, integrating ordinary functions, manipulating polynomials, and integrating ordinary differential and differential-algebraic equations. It is easily extensible and customisable via user-defined functions written in Oc-